DELA AKPALU

Data Scientist | Statistician | Research Analytics

Elkins Park, PA | dela.eak@gmail.com | +1 (336) 978-6189 | dela-ek.github.io
linkedin.com/in/edakpalu

Profile

Statistician and data scientist with graduate training from Wake Forest University and AIMS. Builds reproducible Python, R, and SQL workflows spanning causal inference, experimentation, forecasting, survival analysis, and machine learning. Combines clinical research data experience with modular software pipelines, held-out evaluation, and clear technical communication.

Technical Skills

Programming: Python, R, SQL, C++, Julia, Git, LaTeX
Statistical Methods: Probability, statistical inference, regression, Bayesian statistics, causal inference, A/B testing, difference-in-differences, survival analysis, experimental design, stochastic processes
Machine Learning: scikit-learn, PyTorch, classification, anomaly detection, feature engineering, neural networks, uplift modeling, model selection, precision-recall analysis
Time Series and Data Systems: VAR, recurrent neural networks, nowcasting, temporal validation, Spark, MySQL, REDCap, modular Python pipelines, model persistence, CI
Visualization and Delivery: Matplotlib, Tableau, Power BI, technical reporting, dashboard design, Jira

Selected Projects

Public case studies: dela-ek.github.io. Project repositories are private; code available on request.

Causal Inference and Experimentation *Python, A/B testing, DiD*

Case study

- Developed a 12,000-user randomized experiment and 160-market difference-in-differences study on synthetic data, with market-level uncertainty, event studies, placebo checks, and explicit parallel-trend violations.
- Automated reports and three figures; validated estimators with 12 tests and reproduction checks on Python 3.11/3.12.

Demand Forecasting with Uncertainty *Python, scikit-learn, calibration*

Case study

- Built modular feature, training, calibration, evaluation, and saved-model inference stages; held-out MAE 6.241 versus seasonal-naive 11.878 on synthetic daily demand.
- Reported 84.4% empirical coverage for nominal 90% intervals, documenting undercoverage under serial dependence.

Retention Uplift and Policy Evaluation *Python, S-/T-learners, IPW*

Case study

- Compared heterogeneous-effect models and equal-budget targeting policies in an 18,000-customer synthetic randomized study, with separate train, validation, and test cohorts.
- Evaluated a 30%-capacity policy on 3,600 held-out customers; estimated incremental value 4.40 synthetic units per eligible customer, with pointwise 95% interval [1.16, 7.64].

Customer Churn Survival Analysis *Python, discrete hazards, IPCW*

Case study

- Built person-period hazard modeling, Kaplan–Meier baselines, censoring-aware evaluation, and saved-model scoring for 6,000 synthetic customers; split by customer before period expansion.
- Obtained 12-month IPCW Brier score 0.2199 versus baseline 0.2515 on 1,200 test customers; Harrell concordance 0.662.

Credit-Card Fraud Detection Benchmark *Python, imbalanced learning*

Case study

- Benchmarked supervised and anomaly-detection pipelines; selected random forest with 0.867 average precision and 0.841 F1 on a fixed development split, with class weighting and persisted artifacts.

Time-Series Nowcasting *Python, PyTorch, GRU*

Case study

- Built a modular recurrent-model pipeline with chronological validation and saved preprocessing; validation MAE 0.00962 and separate evaluation MAE 0.01762 on generated market-style data.

Additional projects: Human activity classification (84.4% test accuracy); logistics recommender (Hit Rate@5 9.38%). Both use synthetic data.

Research and Professional Experience

Analytics and Impact Lead **2026–Present**
Greater Faith Deliverance Ministries / Planning Committee / YAYA Tutoring *Project-Based / Volunteer*
- Develop a KPI and analytics framework for program outcomes, volunteer capacity, and operational progress across 21 auxiliaries.
- Support data collection, dashboarding, STEM pilot evaluation, survey design, engagement measurement, and impact reporting.
- Configure Jira workflows to track initiatives, owners, deadlines, blockers, and quarterly progress.

Research Assistant **Jun 2024–May 2025**
Wake Forest University *Winston-Salem, NC*
- Improved REDCap data quality and validation to support reliable storage, retrieval, and analysis of clinical research datasets.
- Automated data-cleaning and validation workflows in R, reducing manual processing and strengthening reproducibility.
- Collaborated with research teams to structure, maintain, and prepare large datasets for statistical analysis and reporting.

Graduate Teaching Assistant **Aug 2023–May 2025**
Wake Forest University *Winston-Salem, NC*
- Taught and supported statistics, inference, exploratory analysis, visualization, and reproducible R/Python workflows.
- Received the 2025 Department of Statistical Sciences Distinguished Teaching Assistant Award.

Mathematics Teacher **Nov 2025–Present**
St. Vincent Academy *Newark, NJ*
- Teach Algebra II, Geometry, and Calculus, emphasizing mathematical reasoning, quantitative fluency, and clear problem-solving.
- Use assessment data to identify learning gaps and develop targeted LaTeX-based notes, assessments, and worked examples.

Teaching Assistant **Sep 2020–Aug 2022**
Kwame Nkrumah University of Science and Technology *Kumasi, Ghana*
- Led instruction in experimental design, algebra, calculus, and statistical methods; mentored students on research and data projects.

Education

M.S., Statistical Science **Aug 2023–May 2025**
Wake Forest University *Winston-Salem, NC*
Causal Inference, Data Mining, Bayesian Statistics, Machine Learning, Survival Analysis

Structured M.Sc., Data Science **Aug 2022–Jun 2023**
African Institute for Mathematical Sciences *Accra, Ghana*
Big Data Analytics, Network Science, High-Dimensional Data Analysis

B.S., Statistics **Sep 2016–Aug 2020**
Kwame Nkrumah University of Science and Technology *Kumasi, Ghana*
Regression Analysis, Demographic Statistics, Stochastic Processes

Publication, Award, and Certification

- **Coauthor, Frontiers in Aging (2025).** "Bone mineral density and turnover response to GLP-1 receptor agonists in older adults with overweight/obesity and prediabetes/type 2 diabetes: a 20-week pilot trial post hoc analysis."
- **Distinguished Teaching Assistant Award,** Wake Forest University Department of Statistical Sciences, 2025.
- **Data Science Certificate,** Wake Forest University, 2025.

Project results above are educational or research experiments and are not investment, medical, financial, legal, or operational advice.